

21 CFR Part 11 Compliant Software – Key Features & Capabilities

Regulatory Compliance & Data Integrity:

- Fully compliant with **21 CFR Part 11** requirements
- Secure **electronic records & electronic signatures (E-records / E-sign)**
- **3-Level E-Signature workflow**: Prepared / Checked / Approved with mandatory reason & remarks
- Complete **paperless operation** with controlled approvals
- Secure, computer-generated **audit trails for every action** (User, Equipment, Alarm, Event, SMS & Email)
- Audit trails include **Who, What, When, Why** (timestamped, non-editable)

User Management & Security Controls:

- **Unlimited user creation & unlimited user groups**
- **Role-based access control** with restricted privileges & rights
- **Alphanumeric password policy** (minimum 8 characters)
- Settable **password ageing & expiry**
- Configurable **auto-logout** feature
- **Invalid login attempt control** with automatic account locking
- Lock / Unlock user facility with full audit trail
- Built-in **biometric integration** (optional)
- Lifetime software **license key** (no forced renewal)

Electronic Signatures & Approval Controls:

- Multi-level electronic signature enforcement
- Mandatory **reason & remarks** for:
 - Alarm & event acknowledgment
 - Configuration updates
- E-signature records remain traceable even after role changes

Equipment & Chamber Management:

- **Multiple equipment / multi-chamber configuration**
- Real-time display of **connected & disconnected chambers**
- **Live equipment status dashboard** with scrolling notifications
- Door open/close operation through software (as applicable)
- Connect / Disconnect chamber control with audit trail
- **Equipment RTC synchronization** (Both Auto & Manual)
- Settable **PID parameters** with secure access

Alarm, Event & Notification Management:

- **Alarm & event recording in seconds**
- Individual sensor-level alarm logging

- Secure **alarm & event acknowledgment** with reason & remark
- Bulk acknowledgment facility with audit trail
- Settable **SMS & Email alerts** for alarms & events
- Scrolling **live alarm & fault notifications** on dashboard
- Backup failure & success notifications on dashboard

Data Logging, Reports & Visualization:

- **Automated background data logging** (no user dependency)
- **Tabular & graphical reports** available
- MKT / MKRH reports with **yearly graphical**
- Real-time and historical data visualization
- E-signature report status & counts visible on dashboard
- Input / Output status display
- 24-hour alarm & event summary view

Backup, Restore & Data Protection:

- **Auto & Manual backup** facility
- Backup success / failure status indication
- Secure data restoration with traceability
- No data overwrite or silent deletion
- Compliance-ready data archival support

System Monitoring & Dashboard Intelligence:

- Centralized dashboard showing:
 - Live chamber status
 - User login & lock status
 - Pending acknowledgments
 - Alarm & event counts
 - Software & equipment health
- Scrolling real-time notifications for critical actions & events

Scalability & Integration:

- Supports **multiple PLC integrations**
- Expandable architecture for future chambers
- No limitation on number of users or chambers